Unit 4 · Outcome 1 · Climate Change

What tips Earth's energy balance?

KK01 VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.

Earth's energy balance

01 Earth's energy balance core idea Almost all of Earth's energy arrives as incoming solar radiation — high-energy shortwave light from the Sun. It leaves again in two ways: some is reflected straight back to space without ever warming anything, and the rest is absorbed, turned into heat, then radiated away as lower-energy longwave (infrared) radiation. Earth's energy balance is simply the comparison of energy in against energy out at the top of the atmosphere. When energy in = energy out , the planet is in equilibrium and its average temperature stays s

Radiative forcing: tipping the scales

02 Radiative forcing: tipping the scales the link word A radiative forcing is any factor that pushes the energy balance away from equilibrium. The direction matters and the exam rewards you for naming it: Positive forcing → warming. The factor adds energy to the system — either letting more sunlight in or stopping heat escaping (e.g. more greenhouse gas). Negative forcing → cooling. The factor removes energy — reflecting sunlight away before it's absorbed (e.g. volcanic haze). A thumb on the scales. Think of energy-in and energy-out as two pans of a b

Factor 1 — Volcanic eruptions

03 Factor 1 — Volcanic eruptions natural · cooling A big explosive eruption blasts sulfur dioxide high into the stratosphere , where it forms a haze of tiny sulfate aerosol droplets. That haze acts like a planet-wide sunshade: it reflects incoming sunlight back to space before it can be absorbed, so less energy enters the system and Earth's surface cools . This is a negative radiative forcing — and a natural one. Mt Pinatubo, 1991 cooling natural ~2 yr effect The Philippines eruption lofted ~20 million tonnes of SO_2 into the stratosphere. Global averag

Factor 2 — Solar variability

04 Factor 2 — Solar variability natural · either way The Sun is the tap, so changes in its output change the whole budget. Solar variability works on two very different clocks: The ~11-year sunspot cycle natural decades small effect The number of sunspots rises and falls about every 11 years, nudging total solar output up and down by roughly 0.1% . More active Sun → a touch more incoming energy → slight warming; quiet Sun → slight cooling. The Maunder Minimum (a 17th-century lull in sunspots) coincided with the cold "Little Ice Age." Milankovitch (o

Factor 3 — Human change to atmospheric gases

05 Factor 3 — Human change to atmospheric gases anthropogenic · warming The third factor is the only anthropogenic one, and it's the dominant driver of the warming measured at Cape Mornay this century. Human activities are changing the gas composition of the atmosphere — raising the concentration of greenhouse gases like carbon dioxide (CO_2), methane (CH_4) and nitrous oxide (N_2O): Burning fossil fuels (coal, oil, gas) for electricity, transport and industry — the largest source of extra CO_2. Land-use change & deforestation — clearing forests releases sto

The Cape Mornay energy desk

06 The Cape Mornay energy desk interactive Here's the station's control desk. Flip each factor and watch the balance beam tip and the thermometer respond. Try to predict the direction before you toggle — that prediction is exactly the exam skill. Live model: sunlight in, reflected + infrared out, and the resulting balance. Volcanic eruption Injects a sulfate-aerosol haze that reflects sunlight away. Brighter Sun Solar output up (active sunspot peak) → more energy in. More greenhouse gas Thicker gas "blanket" traps more outgoing infrared. Energy ba

Sort the forcings

◇ **ACTIVE RECALL**

07 Sort the forcings drag-free classify Tap a card to send it to the bin you think is right — does each factor warm or cool the planet? Get the direction locked before you worry about anything else. Warms (positive forcing) Cools (negative forcing) 0 / 6 placed s

Decode the command word

◈ **COMMAND WORDS**

08 Decode the command word reveal Marks are lost when students answer the wrong kind of question. The command word tells you exactly what the examiner wants. Tap each one. Identify / State Describe Explain Compare Distinguish Evaluate

Six exam traps

⚠ **EXAM TRAPS**

09 Six exam traps avoid these 1 "Volcanoes warm the planet because they release CO_2." The dominant short-term effect is cooling from reflective sulfate aerosols. Volcanic CO_2 is trivial next to human emissions. Write: eruptions cause short-term cooling. 2 Confusing the temperature with the balance. Earth's temperature is set by the difference between energy in and out, not by inflow alone. Always frame it as a balance / imbalance. 3 Saying greenhouse gases "block incoming sunlight." They don't stop sunlight coming in — they slow outgoing infrared from le

P·E·L writing trainer

✎ **P·E·L WRITING**

10 P·E·L writing trainer live feedback High-scoring answers have a shape. Build the habit here: make a Point , back it with Evidence , then Link it to the energy balance. Type below and the meters light up as you hit each part. P — Point Name the factor and its direction: warms or cools. E — Evidence Give the mechanism / example: aerosols, infrared, Pinatubo… L — Link Tie it back to energy in vs out / the balance. Practice prompt: Explain how a major volcanic eruption affects Earth's energy balance. (3 marks) Point 0% — Evidence 0% — Link 0% —

P · POINT answer the verb → **E · EVIDENCE** example / mechanism → **L · LINK** tie back to question

The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Good answer vs bad answer

✗ **WEAK ANSWER**

✗ Weak response 1 / 3 Humans put pollution in the air and this makes the Earth hotter because of global warming . The greenhouse gases block the sun so it gets warmer. Why only 1 mark: vague ("pollution"), circular ("hotter because of global warming"), and a factual error — gases don't block the Sun coming in. No mechanism, no link to the balance.

✓ **STRONG ANSWER & MARKING**

✓ Strong response 3 / 3 Human activities such as bur

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The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Exam-style questions

★ **PRACTICE & SELF-MARK**

12 Exam-style questions 22 marks · self-mark Write your answer first, then reveal. Use the marking guide to score yourself honestly and tap your mark — the dial in the corner keeps your running total.

Quick recap

✦ **RECAP / CHEAT SHEET**

13 Quick recap before you close the tab The KK01 essentials Energy balance = incoming solar (shortwave) vs reflected + outgoing infrared (longwave). In = out → stable temperature. Radiative forcing tips the balance: positive → warming , negative → cooling . Volcanic eruptions — natural, short-term cooling : stratospheric sulfate aerosols reflect sunlight away (Pinatubo 1991, Tambora 1815). Solar variability — natural, either direction: ~11-yr sunspot cycle (small) and Milankovitch orbital cycles (ice ages). Modern trend is flat — can't explain recent war

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.

P · POINT answer the verb → **E · EVIDENCE** example / mechanism → **L · LINK** tie back to question

The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Humans put pollution in the air and this makes the Earth hotter because of global warming . The greenhouse gases block the sun so it gets warmer. Why only 1 mark: vague ("pollution"), circular ("hotter because of global warming"), and a factual error — gases don't block the Sun coming in. No mechanism, no link to the balance.

✓ STRONG / MODEL ANSWER

Human activities such as burning fossil fuels raise the concentration of greenhouse gases like CO_2 . These gases absorb and re-radiate outgoing infrared radiation , so heat escapes to space more slowly. Less energy leaves than enters , creating a positive imbalance, so Earth's surface warms . Mark-earning moves: named human source + named gas (P), correct mechanism on outgoing infrared (E), and an explicit tie to energy out < energy in → warming (L).

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.